Table of values

1 For each of the following linear equations, complete a table of values of the form:

x	1	2	3	4
y				

a $y = 7x$ **b** $y = x + 8$ **c** $y = 2x - 6$ **d** $y = \dfrac{x}{4}$

e $y = 5x + 1$ **f** $y = -3x$ **g** $y = \dfrac{2x}{3}$ **h** $y = 10 - 4x$

2 For each of the following linear equations, complete a table of values of the form:

x	-3	-2	-1	0	1	2	3
y							

a $y = 4x$ **b** $y = x + 3$ **c** $y = 2x + 1$ **d** $y = \dfrac{x}{5}$

e $y = -x + 2$ **f** $y = -2x + 5$ **g** $y = 3x - 4$ **h** $y = 9 - 2x$

Plot points

3 For each of the following equations:

i Complete the table of values.

ii Plot the points from the table on a coordinate plane.

iii State whether the relationship is linear or non-linear.

a $y = 2x$

x	-1	0	1	2
y				

b $y = 3x + 4$

x	-1	0	1	2
y				

c $y = -x + 2$

x	-1	0	1	2
y				

d $y = -4x + 5$

x	-1	1	3	5
y				

e $y = 2x - 4$

x	-2	0	2	4
y				

f $y = -4x - 1$

x	-1	0	1	2
y				

4 Consider the equation $y = \dfrac{3x}{2} - 2$.

a Complete the table of values.

b Plot the points from the table on a coordinate plane.

x	-1	0	1	2
y				

Graph lines

5 Consider the equation $y = -3x$.

a Complete the table of values.

b Use the two points where $x = -1$ and $x = 1$ to sketch the graph of the line.

c Does the point $(0, 0)$ lie on this line?

x	-1	0	1	2
y				

6 Consider the equation $y = 4x + 5$.

a Complete the table of values.

b Use the two points where $x = -1$ and $x = 1$ to sketch the graph of the line.

c Does the point $(2, 13)$ lie on this line?

x	-1	0	1	2
y				

7 For each of the following equations:

i Complete the table of values.

ii Use the two points where $x = -1$ and $x = 1$ to sketch the graph of the line.

x	-1	0	1	2
y				

a $y = -2x + 11$

b $y = \dfrac{5x}{2} - 3$

8 For each of the following linear equations

i Complete the table of values.

ii Use the two axes intercepts, where $x = 0$ and $y = 0$, to sketch the graph of the line.

a $y = 2x - 4$

x	0	1	2	3
y				

b $y = -2x + 4$

x	0	1	2	3
y				

c $y = \dfrac{x}{3} - 2$

x	0	2	4	6
y				

d $y = \dfrac{5x}{3} - 5$

x	0	3	6	9
y				

9 For each of the following linear equations:

i Complete the table of values.

ii Sketch the graph of the line.

a $y = x + 4$

x	-1	0	1	2
y				

b $y = 2x + 4$

x	-1	0	1	2
y				

c $y = -2x - 2$

x	-3	0	3	6
y				

d $y = 2x - 3$

x	-5	0	5	10
y				

e $y = -\dfrac{3x}{2} + 6$

x	-1	0	1	3	5
y					

f $y = \dfrac{x}{5} + 2$

x	-5	0	5	10
y				